

CreditPathAI: Project & Model Report

1. Project Overview

CreditPathAI is an end-to-end credit risk assessment system designed to predict loan defaults. The project transforms raw application data into a predictive model using advanced feature engineering and a Random Forest Classifier.

2. Data Cleaning & Feature Engineering

The pipeline consists of an 11-step process:

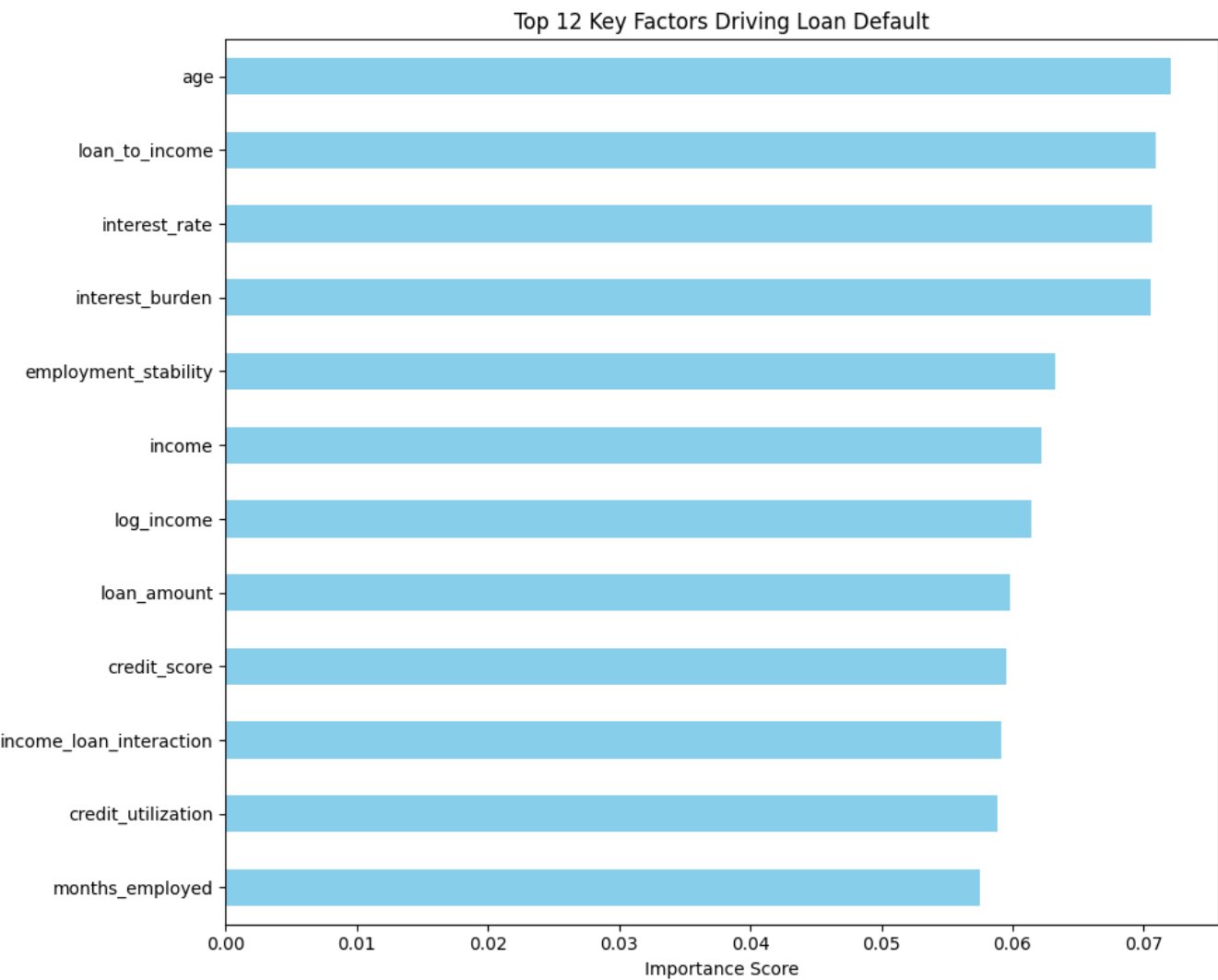
- PostgreSQL extraction with CSV fallback.
- Advanced Feature Engineering: Loan-to-Income, Credit Utilization, Employment Stability, and High DTI Flags.
- Outlier management via Winsorization.
- Standard scaling for all numerical features.
- Categorical encoding for model compatibility.

3. Model Performance & Insights

The Random Forest model was trained on the engineered dataset. Key insights include the importance of credit scores and the newly created interaction features in driving default predictions.

Feature Importance Ranking:

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Confusion Matrix (Predicted vs Actual):

